

CauMed: Causal Graph-Based Reasoning for Medical QA

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Abstract. Large language models (LLMs) in medical question answering (QA) have recently revealed fundamental limitations that hinder their clinical deployment, including an inability to perform causal reasoning beyond pattern matching, the generation of hallucinated responses lacking biomedical grounding, and the failure to quantify reasoning confidence. To address these challenges, we propose CauMed, a causal graph-based framework for clinical reasoning in the medical QA task. CauMed integrates causal interference, biomedical fact validation, and patient-context adaptation. Our model first extracts clinically relevant facts and candidate causal relations from the input questions using a fine-tuned LLaMA-3 model. These relations are then validated via a BioBERT classifier trained in PrimeKG, forming a biomedical causal graph that supports structured inference. To select the final answer, CauMed applies a weight scoring function that combines causal graph support and semantic alignment, ensuring that responses are medically valid, clinically actionable, and contextually coherent. Our experimental results demonstrate that CauMed consistently outperforms baseline methods on the MedQA-US, MedMCQA, and MMLU-Med benchmarks, providing clinically grounded, highly reliable, and trustworthy diagnostic answers.

Keywords: Medical Question Answering · Causal Graph · Biomedical Fact Validation

1 Introduction

Medical reasoning fundamentally differs from other cognitive domains through its reliance on causal inference. Unlike rule-based systems that operate through deterministic logic, clinical decision making requires fluid and analogical reasoning that integrates complex symptomatology with contextual factors specific to the patient [13,11]. When a patient presents with fever, stomach discomfort, and abnormal labs, an accurate diagnosis requires not only listing symptoms but also using systematic causal reasoning to identify etiological correlations and establish pathophysiological connections within temporal constraints. The complexity

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of this reasoning poses fundamental challenges for medical question-answering (QA) systems. Unlike general-purpose systems, developing effective medical QA systems requires moving beyond surface-level semantics toward structured reasoning. Clinical QA must therefore (1) infer causal relationships between symptoms and diseases, (2) express confidence levels for clinical reasoning decisions, and (3) generate safety-critical feedback that meets clinical standards.

One major limitation in a medical QA system is the lack of causal reasoning. Models such as LLaMA2-7B [16], MedAlpaca-7B [5], and PMC-LLaMA [18] are highly proficient at identifying semantic patterns, but they are not designed to reason about the directionality of the hierarchy of cause and effect. For example, a model might learn that “fever” and “infection” frequently appear together but fail to understand that “infection causes fever”. In clinical settings, this misunderstanding can lead to incorrect diagnoses and unsafe treatments. Recent evaluations such as CausalQA [2] and CausalBench [17] show that even advanced LLMs struggle with causal inference, often producing outputs that are logically inconsistent with known medical pathways.

Another key challenge is hallucination, where models produce confident but incorrect answers lacking biomedical grounding. For instance, a model may infer that “smoking causes lung cancer” when asked about chest pain, ignoring other causes such as cardiac conditions. This occurs since LLMs often rely on surface-level co-occurrence rather than validated causal relations. The Fact Finder framework [15] mitigates this issue by retrieving subgraphs from PrimeKG [3] and linearizing them into the model input. While this reduces some hallucinations, converting PrimeKG’s relational structure into flat text removes critical causal dependencies, causing plausible but clinically inconsistent outputs.

Most transformer-based QA systems, such as BioBERT [9], PubMedBERT [4], and ClinicalBERT [1], lack mechanisms to quantify the model’s confidence in its own reasoning. As a result, users cannot distinguish between a highly certain prediction and one generated by ambiguity or a lack of reasoning. This absence of confidence estimation makes it difficult to assess trustworthiness, increases the risk of over-relying on uncertain outputs, and limits safe real-world deployment in clinical decision-support settings.

To address these limitations, we propose CauMed, a causal graph-based framework for clinical QA that combines structured reasoning, and biomedical fact validation. Unlike conventional models that rely solely on language patterns, CauMed constructs causal graph from extracted clinical facts, validates causal edges using a biomedical knowledge base, and performs reasoning over the resulting graph to derive answers. By generating grounded answers in validated causal and biomedical structures, CauMed moves beyond retrieval-style QA and offers a more trustworthy foundation for AI-assisted clinical decision support.

The contributions of this work are summarized as follows:

- We introduced a causal graph-based pipeline for medical question answering that extracts clinical facts, verifies causal relationships, and employs structured multi-step reasoning to enhance answer quality.

- We built a BioBERT-based validation module fine-tuned on PrimeKG, which includes patient-specific context, allowing the system to make more accurate and personalized clinical inferences.
- We developed a hybrid scoring method that combines semantic similarity, causal path strength, and biomedical plausibility, demonstrating consistent improvements over strong transformer-based models in MedQA, MedMCQA, and MMLU-Med.

2 Related Work

Related work is organized into three key areas most relevant to our application: 1) Medical QA; 2) Causal Reasoning in QA; and 3) Biomedical Knowledge Graphs.

2.1 Medical QA

Transformer-based QA systems, such as BioBERT [9], PubMedBERT [4], and ClinicalBERT [1], have advanced understanding of biomedical text, but focus primarily on semantic similarity rather than causal reasoning or patient-specific adaptation. In clinical settings, contextual factors like pregnancy status can drastically alter diagnosis and treatment, meaning that factually correct but de-contextualized answers may still be clinically unsafe. Recent advances in language models have significantly improved performance in medical QA tasks. Notably, LLaMA2-7B [16], MedAlpaca-7B [5], and PMC-LLaMA-7B [18] further improve biomedical inference through instruction tuning and domain adaptation. However, they rely heavily on surface-level correlations and lack structured mechanisms for distinguishing correlation from causation, a key requirement for reliable medical reasoning.

2.2 Causal Reasoning in Clinical QA

Judgement in clinical practice is based on knowledge of symptom and condition interactions elevated to the stage of causal reasoning. Current transformer-based models such as BioBERT [9], PubMedBERT [4], and ClinicalBERT [1] are not engaged in causal inference, focusing exclusively on association. Benchmarks such as CausalQA [2] and CausalBench [17] show that even advanced models, including GPT-4 and LLaMA-3, often fail to identify the correct causal direction. Although resources like CausalBank [10] have begun to introduce causal knowledge, the overwhelming majority store the causal relationships as static and absolute, rather than dynamically reasoning with the knowledge, which further limits the ability to account for the complexity presented by multi-step reasoning in medical question answering.

2.3 Biomedical Knowledge Graphs

Biomedical Knowledge Graphs offer a structured representation of medical entities and relationships, such as disease-symptom-drug interactions, that can enhance the reasoning capabilities of QA systems. Foundational resources, such as

PrimeKG [3], support multi-relational, large-scale reasoning across diverse clinical domains. Although knowledge graphs have been used to improve retrieval or enrich QA inputs, their integration into causal reasoning pipelines remains limited. Most QA systems do not validate generated answers against trusted knowledge graphs, nor do they leverage graph traversal to support multi-hop causal reasoning.

3 The Proposed Approach

To address the challenges in causal reasoning, biomedical validation, and personalization, we introduce CauMed, a modular pipeline designed for medical QA (as shown in Fig. 1). CauMed employs a three-stage process to transform unstructured medical questions into clinically grounded answers through structured causal reasoning. First, a fine-tuned LLaMA-3 model extracts clinical facts and generates candidate causal relationships. In the causal graph construction stage, a BioBERT classifier trained on PrimeKG assigns confidence scores to filter these connections, forming a biomedically grounded causal graph with probabilistic weights. Finally, the contextual answer reasoning stage utilizes a weighted scoring function to integrate causal evidence with semantic similarity via `all-MiniLM-L6-v2` embeddings, ensuring that selections are both medically accurate and contextually coherent. By grounding structured causal inference in established knowledge, CauMed overcomes the limitations of traditional LLMs to provide trustworthy, clinically relevant answers.

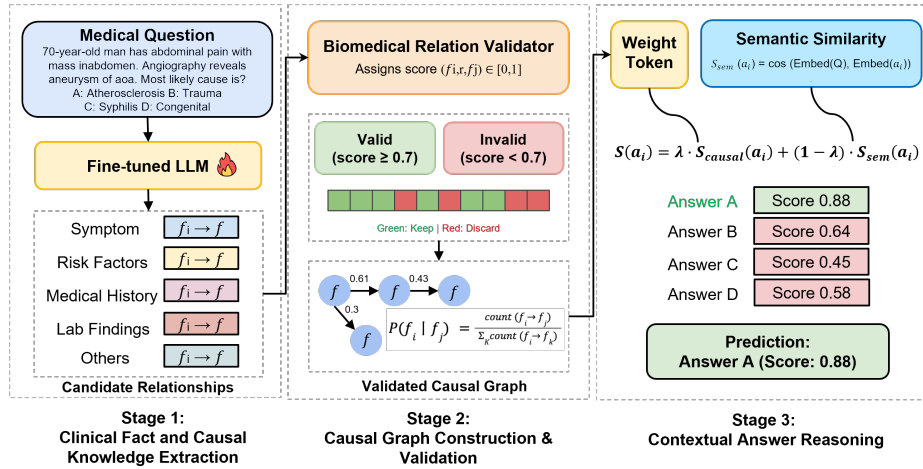


Fig. 1. CauMed Architecture for Medical QA. The framework decomposes the QA process into three stages: (1) Clinical Fact and Causal Knowledge Extraction, (2) Causal Graph Construction and Validation, and (3) Contextual Answer Reasoning.

3.1 Clinical Fact and Causal Knowledge Extraction

The first stage of CauMed converts an unstructured medical question into a structured representation of clinically relevant facts and candidate causal relationships. The question is first processed by a fine-tuned LLaMA-3 model, prompted with custom instructions to extract medical entities from the questions. The extracted entities are semantically categorized into five types: Symptoms, Risk Factors, Medical History, Lab Findings, and Others. Each entity is stored as part of a structured fact set $\mathcal{F} = \{f_1, f_2, \dots, f_k\}$, annotated by category to support interpretability. Once the fact set is extracted, the system jointly infers causal relations among them using a second prompt designed to elicit (Cause $\rightarrow$ Effect) pairs in a clinical context. Output relations are represented in a structured JSON format.

3.2 Causal Graph Construction and Validation

The second stage constructs a directed causal graph from the extracted entities and relationships. The graph is formally defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the nodes $\mathcal{V} = \{f_1, f_2, \dots, f_n\}$ represent extracted medical entities, and the edges $\mathcal{E} = \{(f_i \rightarrow f_j) \mid s(f_i, r, f_j) > \tau\}$ represent validated causal links. To ensure the graph reflects real-world medical knowledge, each candidate’s causal relationship is validated using a BioBERT classifier fine-tuned on the PrimeKG biomedical knowledge graph [3]. The classifier evaluates the plausibility of each relationship $(f_i \rightarrow f_j)$ and assigns a confidence score $s(f_i, r, f_j) \in [0, 1]$. Only relationships with a score exceeding a confidence threshold τ are retained. The threshold $\tau = 0.7$ was selected based on prior work [19] in biomedical relation extraction to balance the inclusion of informative causal links with the exclusion of noisy or speculative associations. This filtering step ensures that only medically validated causal paths are used in the next inference, reducing the risk of LLM hallucination or clinical inaccuracy. For each validated relationship, a causal graph is constructed, where nodes represent medical concepts and edges represent directed causal links. Importantly, CauMed leverages transition frequency statistics to assign probabilistic weights to edges based on how often a given cause leads to a specific effect across all observed data. The transition probability from a source node f_i to a target node f_j is defined as:

$$P(f_j \mid f_i) = \frac{\text{count}(f_i \rightarrow f_j)}{\sum_k \text{count}(f_i \rightarrow f_k)} \quad (1)$$

Where, f_i is a source node (e.g., a disease or condition), f_j is a target node (e.g., a symptom or finding), $\text{count}(f_i \rightarrow f_j)$ denotes the number of times the system observes or infers this specific causal relation across all processed question-answer pairs in the dataset or current query, $\sum_k \text{count}(f_i \rightarrow f_k)$ is the total number of causal transitions originating from f_i summing over all its outgoing edges. This formulation captures the relative frequency of each causal link, reflecting how often a symptom or finding (e.g., “fever”) is associated with a given

diagnosis (e.g., “appendicitis”). To prevent data leakage, these counts are derived strictly from the relations extracted within the current query context or from prior statistics in the training corpus, ensuring that the model does not look ahead at evaluation labels.

3.3 Contextual Answer Reasoning

The final stage integrates causal graph-based reasoning with semantic alignment to select the most appropriate answer. CauMed first evaluates biomedical plausibility using the validated causal graph from Stage 2, where each directed edge carries a weight token, the transition probability $P(f_j | f_i)$. These tokens represent the strength of each causal link and are multiplied along a path to estimate its likelihood. For each answer choice a_i , CauMed searches for the most plausible causal chain connecting the extracted clinical facts to that answer. An answer a_i is considered to involve a clinical fact f_k if the underlying concept of f_k is semantically expressed in the answer, either explicitly or through lexical or conceptual paraphrasing. The causal support is then defined as:

$$S_{\text{causal}}(a_i) = \sum_{(f_j \rightarrow f_k) \in \mathcal{E}} P(f_k | f_j) \cdot \mathbf{1}[a_i \text{ involves } f_k] \quad (2)$$

Next, semantic similarity scoring ensures that the selected answer is phrased in a way that is contextually relevant to the question Q . We compute the cosine similarity between the embedding of the question and each answer choice using the `all-MiniLM-L6-v2` model from SentenceTransformers:

$$S_{\text{sem}}(a_i) = \cos(\text{Embed}(Q), \text{Embed}(a_i)) \quad (3)$$

The final prediction score for each answer combines its causal support and semantic alignment using a weighted formula:

$$S(a_i) = \lambda \cdot S_{\text{causal}}(a_i) + (1 - \lambda) \cdot S_{\text{sem}}(a_i) \quad (4)$$

Here, $\lambda \in [0, 1]$ is a tunable parameter that controls the relative importance of causal evidence versus semantic similarity. A higher λ places more emphasis on biomedical causality, while a lower value gives greater weight to linguistic coherence. The answer choice with the highest final score is selected as the system’s prediction. This scoring mechanism ensures that the final output is both biomedically validated and linguistically coherent, aligning closely with human clinical reasoning.

4 Experiments

We evaluate the CauMed framework for multiple-choice clinical question answering by comparing its performance against state-of-the-art baseline models across benchmark datasets.

4.1 Datasets

Experiments are conducted on three established medical QA datasets. The first, MedQA-US [7], contains USMLE-style board exam questions authored by physicians. These questions are known for requiring multi-symptom reasoning and contain plausible distractors. We sample 2,000 questions from the development set for evaluation. The second dataset, MedMCQA [12], is a large-scale QA corpus sourced from Indian medical entrance exams, covering diagnosis, treatment, and factual recall. A 2,000-question subset is also sampled for consistency. The third dataset, MMLU-Med [6], is a medical domain subset of the Massive Multitask Language Understanding benchmark, focusing on evaluating general medical knowledge in a five-choice multiple-choice format. Each dataset presents unique challenges: MedQA favors complex reasoning and distractor disambiguation, MedMCQA emphasizes factual breadth and topic coverage, and MMLU-Med tests foundational biomedical knowledge across academic subfields.

4.2 Baselines

We consider two main groups of baseline models:

Biomedical Transformer Models: These include BioBERT [9], PubMedBERT [4], and ClinicalBERT [1]. All three are encoder-only BERT variants pre-trained on biomedical corpora such as PubMed [14] and MIMIC-III [8] clinical notes. These models are fine-tuned for multiple-choice classification and serve as strong baselines for biomedical understanding without external reasoning tools.

Large Language Models (LLMs): These include LLaMA2-7B [16], PMC-LLaMA-7B [18], and MedAlpaca-7B [5]. MedAlpaca is a domain-specific variant of LLaMA2, fine-tuned on biomedical corpora, whereas PMC-LLaMA incorporates content from PubMed Central. All LLMs are used in zero-shot multiple-choice settings without Chain-of-Thought prompting or retrieval augmentation.

4.3 Evaluation Setup

The CauMed framework begins with entity and relation extraction using a domain-prompted LLaMA-3 model. To ensure high-fidelity extraction, we utilized a zero-shot prompting strategy with custom instructions designed to categorize entities into five distinct clinical types: Symptoms, Risk Factors, Medical History, Lab Findings, and Others. This approach allows the system to remain modular and adaptable to various clinical datasets without the need for extensive task-specific fine-tuning. The extracted candidate relationships are passed through a plausibility checker, a BioBERT classifier fine-tuned on PrimeKG, where triplets are filtered based on a confidence threshold of $\tau = 0.7$. After constructing a causal graph, CauMed computes semantic similarity between each answer choice and relevant parts of the graph using the `all-MiniLM-L6-v2` model from SentenceTransformers. For final answer selection, CauMed assigns a weighted score that combines causal graph evidence (weight $\lambda = 0.6$) and semantic similarity (weight $(1 - \lambda) = 0.4$). The highest-scoring answer is returned as the final prediction.

All models are evaluated in a multiple-choice setting, where each input consists of a clinical question accompanied by four to five candidate answer options, with a single correct answer. Models are tasked with selecting the correct answer from the set. Accuracy is computed as the proportion of questions for which the model selects the correct answer. All experiments were executed on a workstation equipped with an NVIDIA RTX A5000 (24 GB) GPU and an AMD EPYC 7302 16-core processor.

4.4 Results and Analysis

Table 1 presents the accuracy of CauMed compared to six baseline models across the MedQA-US, MMLU-Med, and MedMCQA datasets. The results offer insights into the reasoning capabilities and domain alignment of different model families, namely: (i) biomedical language models based on BERT architectures and (ii) decoder-only LLMs such as LLaMA2-7B and its biomedical variants. In a zero-shot setting without chain-of-thought prompting or external access, BioBERT, PubMedBERT, and ClinicalBERT consistently underperform, with accuracy scores ranging from 0.190 to 0.210. While these encoder-based architectures excel in tasks like biomedical entity recognition, they struggle with the complex, symptom-based, and causal reasoning required for multiple-choice medical QA. Although their slightly higher performance on MMLU-Med suggests a capacity for basic factual recall, their inherent architectural limitations in comparing and reasoning across multiple answer choices significantly restrict their effectiveness in clinical QA tasks.

Table 1. Accuracy (Acc.) and Standard Deviation (Std.) of CauMed and baseline models on three clinical QA benchmarks under a zero-shot multiple-choice setting. The highest accuracy is in bold; the second-highest is underlined.

Method	MedQA-US	MedMCQA	MMLU-Med
	Acc. (std.)	Acc. (std.)	Acc. (std.)
BioBERT	0.196(.012)	0.195(.011)	0.198(.013)
PubMedBERT	0.200(.013)	0.210(.014)	0.190(.012)
ClinicalBERT	0.191(.011)	0.190(.012)	0.191(.012)
LLaMA2-7B	0.281(.014)	0.376(.016)	0.508(.014)
MedAlpaca-7B	0.401(.014)	0.493(.015)	0.600(.014)
PMC-LLaMA-7B	0.247(.011)	0.207(.012)	0.446(.027)
CauMed	0.521(.022)	0.527(.022)	<u>0.574(.023)</u>

Decoder-based language models, including LLaMA2-7B, MedAlpaca-7B, and PMC-LLaMA-7B, show significant improvement over biomedical transformers. LLaMA2-7B achieves 0.508 accuracy on MMLU-Med, benefiting from its strong language modeling capabilities. However, its performance drops sharply on reasoning-based benchmarks, such as MedQA-US (0.281), suggesting that while it can re-

trieve factual knowledge, it struggles with the multi-step inference required for clinical decision-making. MedAlpaca-7B, which is fine-tuned on biomedical texts, shows better consistency, scoring 0.401 on MedQA-US and 0.600 on MMLU-Med. This highlights the value of domain adaptation in improving medical content understanding. In contrast, PMC-LLaMA-7B yields unstable results, achieving only 0.207 on MedMCQA, despite demonstrating stronger performance on MMLU-Med. This may be due to overfitting on academic writing styles rather than clinical reasoning structure, leading to poor generalization.

CauMed achieves the highest accuracy on MedQA-US (0.521) and MedMCQA (0.527), alongside the second-highest performance on MMLU-Med (0.574), outperforming the strongest LLM baseline (MedAlpaca-7B) by +12.0% and +3.4%, respectively. These results indicate that CauMed not only matches LLMs in factual recall but provides a significant advantage in structured reasoning through its modular design, which integrates biomedical entity extraction, causal validation, and semantic similarity scoring. By tracing logical connections between symptoms, diagnoses, and treatments via a causal reasoning graph, CauMed handles ambiguous questions more effectively than LLMs relying solely on parametric memory, allowing it to reject distractors and provide interpretable outputs. Furthermore, CauMed maintains consistent stability across datasets, unlike models such as PMC-LLaMA-7B, which fluctuate between 0.207 and 0.346. This stability, achieved without fine-tuning or prompt engineering, supports the idea that integrating clinical knowledge through causal graph reasoning enhances both accuracy and robustness in medical QA.

4.5 Ablation Analysis

To assess the individual contributions of each component, we perform the ablation studies across three clinical QA benchmarks: MedQA-US, MedMCQA, and MMLU-Med (as shown in Table 2). As the cornerstone of CauMed’s reasoning architecture, the causal graph enables the simulation of clinical cause-and-effect relationships, without which performance degrades by 20.9% in MedQA-US, 16.5% in MedMCQA, and 17.8% in MMLU-Med. Experiments with randomized

Table 2. Accuracy (Acc.) and Standard Deviation (Std.) of CauMed and its ablated variants on three clinical QA benchmarks under a zero-shot multiple-choice setting. Highest accuracy is in bold; second-highest is underlined.

Method	MedQA-US	MedMCQA	MMLU-Med
	Acc. (std.)	Acc. (std.)	Acc. (std.)
w/o BioBERT Classifier	0.411(.021)	0.494(.022)	0.545(.023)
w/o Causal Graph	0.312(.018)	0.362(.019)	0.396(.024)
Randomized Graph	0.308(.018)	0.385(.020)	0.414(.022)
w/o Graph Weights	0.300(.020)	0.381(.020)	0.420(.022)
w/o Semantic Scoring	0.303(.018)	0.358(.019)	0.401(.022)
CauMed	0.521(.022)	0.527(.022)	0.574(.023)

graph connections reveal comparable deterioration patterns (21.3%, 14.2%, and 16.0%, respectively), confirming that the specific graph structure is as crucial as its existence; without it, random associations sever logical connections between symptoms, risk variables, and diagnoses. Furthermore, the graph weighting mechanism proves essential for prioritizing clinically important links, as its removal reduces accuracy by 22.1%, 14.6%, and 15.4% by allowing weaker, less relevant connections to yield less trustworthy outcomes.

Similarly, semantic scoring is also vital for ensuring predictions are contextually and causally compatible with inquiries, as its removal leads to accuracy declines of 21.8%, 16.9%, and 17.3% by causing the system to overlook equivalent semantic content across different linguistic phrasings. While the BioBERT classifier shows a more moderate individual impact (improving performance by 11.0%, 3.0%, and 2.9%), it remains a crucial quality control mechanism for filtering inaccurate medical associations that might mislead the reasoning process. These results are particularly notable in MMLU-Med, which exhibits the largest decline due to its comprehensive coverage of diverse medical issues. Ultimately, consistent performance across all benchmarks, with standard deviations below 0.024, demonstrates that integrating causal reasoning with semantic alignment and maintaining therapeutically valid linkages are essential for capturing clinically correct responses in challenging medical QA tasks.

4.6 Case Study

To illustrate the end-to-end reasoning capability of CauMed, we present a representative case study from the MedQA-US dataset (summarized in Tables 3 and 4). The question concerns a “35-year-old woman presenting with heat intolerance, palpitations, weight loss, lid retraction, moist palms, and suppressed TSH with elevated T3/T4”. The answer choices are: A) Hyperthyroidism, B) Hypothyroidism, C) Thyroid cancer, and D) Cushing syndrome. Using this validated graph, CauMed computes causal support and semantic similarity scores for each answer option. The results show that Hyperthyroidism achieves the highest score, reflecting strong causal alignment with the extracted clinical evidence.

Table 3. Key Outputs of Stage 1 (Clinical Fact Extraction) and Stage 2 (Triplet Generation and Validation) in the Pipeline

Stage	Pipeline Component	Example Output
Stage 1	Clinical Facts	Heat intolerance, Palpitations, Lid retraction, Weight loss, Low TSH, High T3/T4
	Candidate Triplets	Hyperthyroidism → Palpitations; Hyperthyroidism → Lid retraction; Anemia → Cancer
Stage 2	Validated Triplets	Hyperthyroidism → Heat intolerance; Hyperthyroidism → Moist palms; Hyperthyroidism → Palpitations

Table 4. Causal and Semantic Scores for Case Study Answer Choices

Answer a_i	S_{causal}	S_{sem}	$S(a_i)$	
A. Hyperthyroidism	1.00	0.29	0.71	✓
B. Hypothyroidism	0.25	0.22	0.24	✗
C. Thyroid cancer	0.25	0.15	0.21	✗
D. Cushing syndrome	0.00	0.11	0.04	✗

5 Conclusion

In this work, we propose CauMed, a novel framework for medical question answering that incorporates structured causal reasoning to enhance interpretability and reliability. By constructing a causal graph grounded in biomedical knowledge, CauMed outperforms both encoder-based and decoder-only models across the MedQA-US, MMLU-med, and MedMCQA benchmarks, demonstrating particular strength in reasoning-intensive clinical tasks. Operating as a fully modular, end-to-end pipeline, the system offers a significant step toward trustworthy, explainable, and clinically aligned AI. While this study focuses on multiple-choice formats to provide objective comparisons, we acknowledge that such benchmarks only partially reflect the complexity of real-world clinical synthesis. Future work will explore free-text generation to provide more explainable diagnostic outputs. Additionally, to address long-tail knowledge gaps regarding rare diseases, we plan to integrate Retrieval-Augmented Generation (RAG) to couple our causal reasoning with the latest biomedical literature. Based on RAG, we can also incorporate multilingual datasets, evaluate against new medical foundation models, and explore knowledge-graph-native generative models to reduce LLM dependency and boost computational efficiency.

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